PPE COMPLIANCE DETECTION SYSTEM

Multi-Class Safety Equipment Detection using YOLOv8

MAICEN-1125 | M4 Unit 4 | Group 5

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THE CHALLENGE

5,000+

workplace fatalities annually (US)

84%

of head injuries occur without hard hats

60%

of construction deaths are PPE-preventable

OUR SOLUTION

Automated 4-class PPE detection system using YOLOv8 computer vision to monitor construction site compliance in real-time.

Detects: Helmets, Bare Heads, Safety Vests, and Goggles — with a safety-first threshold of 0.25 to maximize recall.

4-CLASS DETECTION SYSTEM

0

Helmet

XML Annotations

Hard hat detection

16,195

labels

1

Head

XML Annotations

Bare head (violation)

4,850

labels

2

SafetyVest

Pseudo-Labeled

Hi-vis vest detection

3,206

labels

3

Goggles

Pseudo-Labeled

Safety eyewear

4,209

labels

Total: 28,460 labels across 4,250 images

KEY INNOVATION: PSEUDO-LABELING

THE PROBLEM

Original XML annotations only contain Helmet and Head classes. No labels exist for Safety Vests or Goggles in the dataset.

OUR SOLUTION

Generate pseudo-labels at runtime using computer vision techniques — no manual annotation required.

Safety Vest Detection

HSV Color Segmentation

Detects high-visibility yellow/orange vests within person bounding boxes using color thresholds in HSV space (threshold: 0.15).

3,206 vests detected

Goggles Detection

Canny Edge Detection

Analyzes eye region within detected heads for circular edge patterns characteristic of safety goggles (threshold: 0.45).

4,209 goggles detected

MODEL ARCHITECTURE & TRAINING

YOLOv8m Architecture

- Pre-trained on COCO (80 classes)
- 25.9M parameters
- Fine-tuned for 4-class PPE detection
- Anchor-free detection head
- CSPDarknet53 backbone with C2f modules
- Multi-scale feature fusion (P3-P5)

Training Configuration

Epochs	**30**
Batch Size	**16**
Image Size	**640 x 640**
Optimizer	**Auto (SGD)**
Learning Rate	**0.01 (cosine decay)**
Patience	**7 (early stopping)**
Augmentation	**Mosaic + RandAugment**
Confidence	**0.25**
GPU	**RTX 4000 Ada (12GB)**
Dataset	**4,696 train / 750 val**

MODEL PERFORMANCE

74.2%	47.0%	73.5%	72.9%
mAP50	mAP50-95	Precision	Recall

Class	Precision	Recall	F1 Score	mAP50	Source
Helmet	0.923	0.941	**0.932**	**0.970**	XML
Head	0.880	0.919	**0.899**	**0.930**	XML
SafetyVest	0.553	0.778	**0.647**	**0.714**	Pseudo
Goggles	0.474	0.354	**0.405**	**0.332**	Pseudo

XML-annotated classes (Helmet, Head) achieve strong F1 > 0.89. Pseudo-labeled classes show promising results given zero manual annotation.

SAFETY-FIRST THRESHOLD ANALYSIS

FALSE NEGATIVE

Missed bare head = CATASTROPHIC

Worker injury or death

Legal liability (millions)

Site shutdown

Reputation damage

FALSE POSITIVE

False alarm = LOW COST

Brief manual check (~30 sec)

Minor productivity loss

No safety impact

Easily dismissed

Confidence threshold set to 0.25 — deliberately low to maximize RECALL and minimize missed safety violations. In safety-critical systems, it is always better to have false alarms than missed detections.

CONCLUSIONS & FUTURE WORK

KEY ACHIEVEMENTS

- Built a 4-class PPE detection system with zero manual annotation for 2 classes
- Achieved mAP50 of 97.0% (Helmet) and 93.0% (Head) on XML-annotated classes
- Demonstrated pseudo-labeling innovation: generated 7,415 labels for SafetyVest and Goggles
- Safety-first design: low threshold (0.25) prioritizes recall over precision

FUTURE WORK

- Manual annotation of SafetyVest/Goggles for improved pseudo-label quality
- Real-time video stream integration for live site monitoring
- Edge deployment on NVIDIA Jetson for on-site processing
- Multi-camera zone tracking with automated compliance reporting